



SPARQL **(SPARQL Protocol and RDF Query Language)**

Knowledge Base and Reasoning

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SPARQL

- SPARQL Protocol and RDF Query Language
- **SPARQL is the RDF query language.**
- It is a semantic query language for semantic data, which can retrieve data stored in RDF format.
- Like SQL for the relational data model, **SPARQL is a matching query language.**
- SPARQL covers the abilities to **query all RDF serializations including Named Graph**. Although Named Graph proposed its own query language named TRIQL, but it is not a part of the W3C recommendations



Jena Apache

- <https://jena.apache.org>
- Semantic Web framework
- It is based on Java
- RDF stores
- Detail: <https://www.w3.org/TR/sparql11-query/>



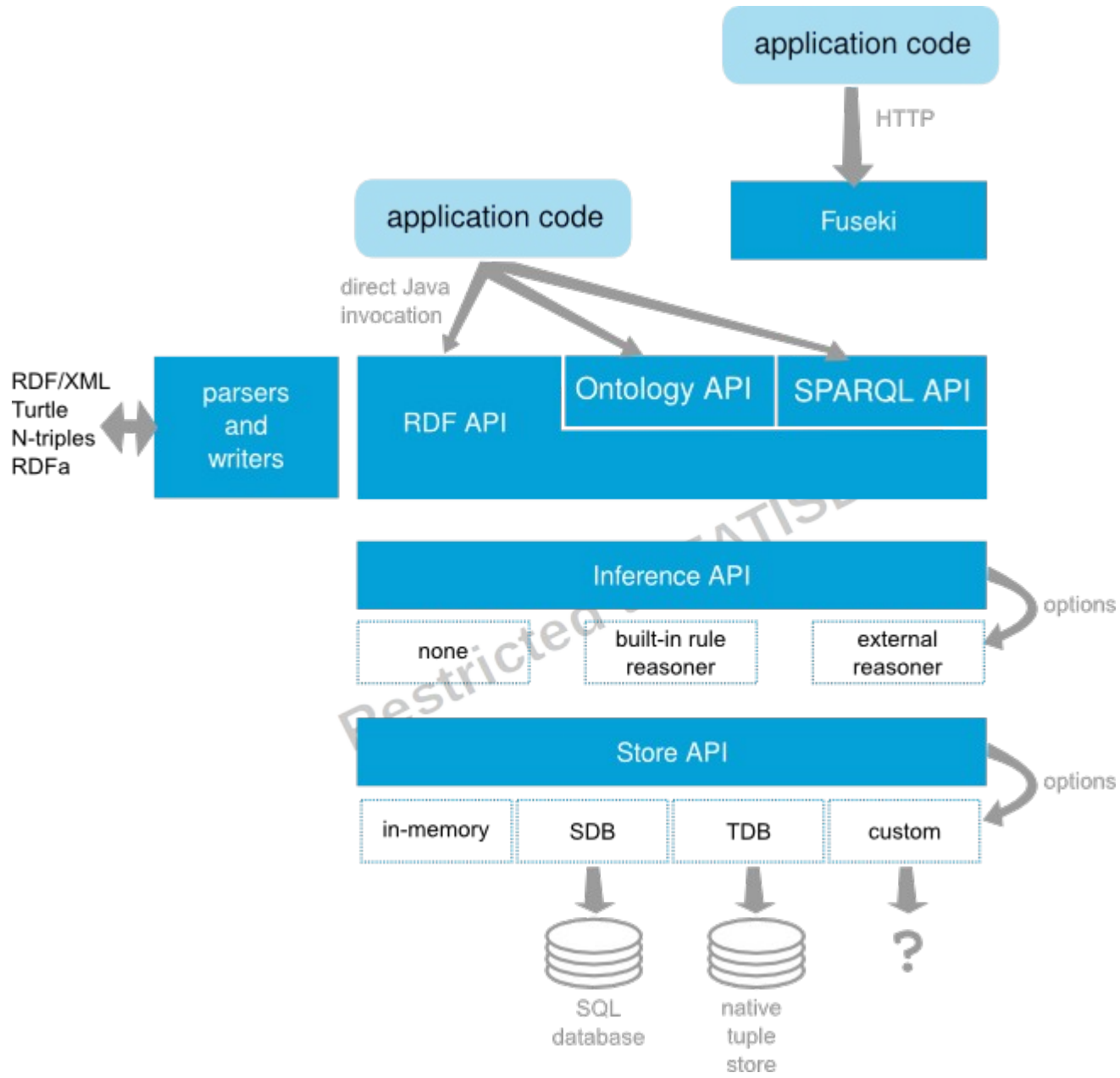
Triple Store

- 4store, AllegroGraph RDF Store, Mobi, Mulgara Semantic Store, Ontop, Ontopic Studio, OpenLink Virtuoso, Oracle Spatial and Graph 19c, GraphDB, RDFLib, RDFox, Redland RDF Application Framework, Sesame, Stardog
- etc



Type

- SELECT → finding
- DESCRIBE → description
- CONSTRUCT → add/ create, mengganti
- ASK → true/ false
- Element
 - S: IRI, Blank node
 - P: IRI
 - O: IRI, Blank node, Literal





Select

- SELECT ?s ?p ?o
WHERE { ?s ?p ?o }

SELECT ?s ?p ?o
WHERE {GRAPH ?g { ?s ?p ?o } }

s	p	o
<http://dbpedia.org/resource/Netherlands>	<http://ontosem.org/#colonize>	<http://worldbank.org/Indonesia>

s	p	o
<http://dbpedia.org/resource/Netherlands>	<http://ontosem.org/#colonize>	"Indonesia"^^<http://www.w3.org/2001/XMLSchema#literal>

- In Turtle, it starts with Prefix

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?subject ?object

WHERE { ?subject rdfs:subClassOf ?object }



Ex: Familiy

- List class/konsep
- List class yg punya subclass
- List class dan subclassnya
- List individu woman
- List individu man, yang punya anak



Describe

- All about a specific resource

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
```

```
DESCRIBE ?subject ?predicate ?object
```

```
WHERE { ?subject ?predicate ?object }
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
```

```
DESCRIBE :Woman
```



Ask

- True/False → result obtaining

PREFIX rdf: <<http://www.w3.org/1999/02/22-rdf-syntax-ns#>>

PREFIX owl: <<http://www.w3.org/2002/07/owl#>>

PREFIX rdfs: <<http://www.w3.org/2000/01/rdf-schema#>>

PREFIX xsd: <<http://www.w3.org/2001/XMLSchema#>>

ASK { ?subject hasWife ?object }

- Ex:
 - Apakah ada individual Bill
 - Apakah Bill beristri Mary



Construct

- Matching triple berdasar template

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

CONSTRUCT ?object hasHusband ?subject
WHERE { ?subject hasWife ?object }

- Ex:

- Buat triple ... jika



Insert

- Update the triple, similar with CONSTRUCT but it can update triple

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

INSERT {?object hasHusband ?subject}

WHERE { ?subject hasWife ?object }



Delete

- Deleting triple, similar with CONSTRUCT but it can delete triple
- ```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

DELETE {?object hasHusband ?subject}
WHERE { ?subject hasWife ?object }
```



# Modification

- DISTINCT
- ORDER BY
- LIMIT n
- OFFSET m
- Filtering, Remove duplicate
- Aggregation
- Grouping, Subqueries
- Union
- Property Paths

Restricted to FATISDA



## **\* , Limit**

- Limitation resulted query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
```

```
SELECT *
WHERE { ?subject rdf:type owl:Class }
LIMIT 10
```



# Order By

- Sorting

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT \*

WHERE { ?subject rdf:type owl:Class }

ORDER BY ?subject

LIMIT 10

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT \*

WHERE { ?subject rdf:type owl:Class }

ORDER BY desc (?subject)

LIMIT 10



# Group By, Aggregation

- For grouping, normally works with aggregation function

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
```

```
SELECT ?subject (count(?subject) as ?count)
```

```
WHERE { ?subject rdf:type owl:Class }
```

```
GROUP BY ?subject
```



# Operator, FILTER

- Operator: Arithmetic, Logic, String, Conditionals, dll
- FILTER, to handle condition of where  
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX owl: <http://www.w3.org/2002/07/owl#>  
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>  
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>  
  
SELECT \*  
{ ?s ex:hasAge ?o  
FILTER (?o > 40)  
}



# Jena Apache

- Exercise 1 :

@prefix dbpedia:<http://dbpedia.org/resource/> .

@prefix wb:<http://worldbank.org/> .

@prefix osem:<http://ontosem.org/#> .

@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

@prefix : <http://www.example.org/#> .

dbpedia:Netherlands osem:colonize wb:Indonesia .

- Netherlands colonized Indonesia from 1595 until 1942. Next, Japan colonized Indonesia from 1942 until 1945. Indonesia's independence day is August 17, 1945. It was Friday. Soekarno and Muhammad Hatta were the declarators of the independence.



- Queries:
  - Who was the declarator of Indonesia's independence?
  - When is Indonesia's independence day?
- Continue ...

The location was at Soekarno's house. The address was at Jl. Pegangsaan Timur no. 56, Jakarta. The first location plan was at Lapangan Banteng. Soekarno was sick on that day.

- Rewrite it in N-Quads



Write other queries below:

- Who colonized Indonesia?
- Where was the address of Soekarno's house?
- Where was the location of Indonesia's independence?
- Who colonized Indonesia in 1910?

→ Also write the SPARQL for N-Quads